

CS1303 Full Stack Development — Assignment 3 Report

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Course: CS1303 Full Stack Development

Assignment: Assignment 3 (Backend Integration)

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GitHub Repo: <https://github.com/HarshaVardhan31012007/Portfolio>

Google Drive Link: [PASTE_YOUR_GOOGLE_DRIVE_LINK_HERE]

1. Executive Summary

This project extends the React Portfolio Website built in Assignment 2 by integrating a fully functional **Node.js / Express REST API backend** located in the `/server` directory. Hardcoded static projects and client-side contact submission have been transitioned to dynamic backend API endpoints with JSON file persistence, server-side validation, CORS, and comprehensive error handling.

2. Backend Tasks Evaluation Matrix (B1–B7)

Task	Endpoint / Resource	Method	Status	Evaluation Specification & Implementation Details
B1	<code>/</code>	GET	Complete	Health check route returning HTTP 200 <code>{"status": "ok"}</code> . Server port loaded from <code>PORT</code> in <code>.env</code> .
B2	<code>/api/projects</code>	GET	Complete	Serves array of 4 projects from <code>server/data/projects.json</code> retaining all props expected by <code>ProjectCard</code> .
B3	<code>/api/projects/:id</code>	GET	Complete	Returns project matching valid ID with HTTP 200, or HTTP 404 JSON <code>{"error": "Project not found"}</code> for invalid IDs.
B4	<code>/api/contact</code>	POST	Complete	Validates <code>name</code> , <code>email</code> (with "@"), and <code>message</code> . Returns 400 on error, or 201 on valid submission and persists to <code>contacts.json</code> .
B5	<code>/api/contact</code>	GET	Complete	Open verification endpoint returning all stored contact submissions in JSON.
B6	<code>/*</code> (Catch-all)	ALL	Complete	Centralized catch-all 404 middleware for undefined routes and global Express error-handling middleware (<code>err</code> , <code>req</code> , <code>res</code> , <code>next</code>).
B7	CORS & Env	-	Complete	Configured <code>cors</code> for <code>http://localhost:5173</code> . Provided <code>.env.example</code> with zero secrets committed.

3. Frontend Integration Tasks (F1–F4)

Task	Component	Status	Implementation Details
F1	src/pages/Projects.jsx	Complete	Fetched data via <code>useEffect</code> calling <code>GET /api/projects</code> . Displays <code>LoadingSpinner</code> while fetching.
F2	Projects.jsx & Home.jsx	Complete	Catches network errors gracefully when backend is offline, rendering a connection error banner with a "Retry Connection" button.
F3	src/pages/ProjectDetail.jsx	Complete	Dynamic parameter <code>useParams()</code> triggers <code>GET /api/projects/:id</code> . Supports direct deep links and renders 404 page for invalid IDs.
F4	src/components/ContactForm.jsx	Complete	Submits form via <code>POST /api/contact</code> . Displays green success banner on 201 or renders server-side validation error message on 400 rejection.

4. Execution & Startup Commands

The application runs using two documented commands:

```
# 1. Start Express Backend Server (http://localhost:5000)
npm run server

# 2. Start React Frontend Dev Server (http://localhost:5173)
npm run dev
```

5. Verification Suite & cURL Test Commands

```
# B1: Health Check
curl -X GET http://localhost:5000/

# B2: Get All Projects
curl -X GET http://localhost:5000/api/projects

# B3: Get Single Project (Valid ID)
curl -X GET http://localhost:5000/api/projects/stayverse

# B3: Get Single Project (404 Non-existent ID)
curl -X GET http://localhost:5000/api/projects/invalid-id

# B4: Valid Contact Submission (201 Created)
curl -X POST http://localhost:5000/api/contact \
  -H "Content-Type: application/json" \
  -d '{"name":"Alex Turner","email":"alex@example.com","message":"Great portfolio! Looking forward to connecting"}'

# B4: Invalid Email Submission (400 Bad Request)
curl -X POST http://localhost:5000/api/contact \
  -H "Content-Type: application/json" \
  -d '{"name":"Alex Turner","email":"invalidemail","message":"Short message"}'

# B5: List All Contact Submissions (Open Verification Endpoint)
curl -X GET http://localhost:5000/api/contact

# B6: Catch-All 404 Route Test
curl -X GET http://localhost:5000/api/doesnotexist
```

Task B5 Open Endpoint Notice: As required by the assignment specification, GET /api/contact is an open endpoint without authentication for evaluation and testing purposes.